

Customer Purchasing Behavior Analysis

1. Problem Statement:

The objective of this project is to analyze customer purchasing behavior by reducing high-dimensional purchasing data into a two-dimensional space using dimensionality reduction techniques. The project also aims to identify meaningful customer segments using clustering techniques for better marketing strategies and personalized recommendations.

2. Objective:

The objectives of this project are:

- To visualize customer purchasing behavior in a 2D space using Principal Component Analysis (PCA).
 - To discover hidden patterns and nonlinear relationships using t-Distributed Stochastic Neighbor Embedding (t-SNE).
 - To segment customers into different groups using K-Means Clustering.
 - To compare the effectiveness of PCA, t-SNE, and K-Means visualization techniques.
 - To provide business insights that can support targeted marketing and customer segmentation.
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3. Dataset Information

Dataset Name

Synthetic Customer Purchasing Dataset

Source Link

Generated synthetically for demonstration purposes (No external dataset used).

Task Type:

Unsupervised Learning

Dataset Details:

- Number of Customers : 500
 - Number of Features : 10
 - Features : Purchase amount across ten product categories
 - Preprocessing : StandardScaler (Z-score Normalization)
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4. Implementation

Algorithm 1: Principal Component Analysis (PCA)

Code:

```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

print("Explained variance ratio:", pca.explained_variance_ratio_)
print("Total explained variance:", sum(pca.explained_variance_ratio_))

plt.figure(figsize=(10, 8))
plt.scatter(X_pca[:, 0], X_pca[:, 1], alpha=0.7, c='steelblue', edgecolor='k')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.title('PCA Projection of Customer Purchasing Data')
plt.grid(True)
plt.show()
```

Confusion Matrix

Not Applicable

Interpretation:

PCA is an unsupervised dimensionality reduction algorithm. Since there are no actual class labels available, a confusion matrix cannot be generated.

PCA projects the original high-dimensional purchasing data into two principal components while preserving most of the variance.

Graph:



✓ PCA Projection of Customer Purchasing Data

Graph Interpretation

The PCA visualization reduces the original ten-dimensional dataset into two principal components.

- Principal Component 1 captures approximately **89.48%** of the total variance.
- Principal Component 2 captures approximately **4.31%** of the variance.
- Together they explain approximately **93.79%** of the total dataset variance.

The graph shows that customer purchasing behavior mainly differs based on overall spending patterns.

Metrics

Metric	Value
Accuracy	Not Applicable
Precision	Not Applicable
Recall	Not Applicable
F1-Score	Not Applicable
AUC-ROC	Not Applicable
MSE	Not Applicable
RMSE	Not Applicable
MAE	Not Applicable
R ²	Not Applicable

Algorithm 2: t-Distributed Stochastic Neighbor Embedding (t-SNE)

Code:

```
tsne = TSNE(n_components=2, perplexity=30, random_state=42, n_iter=1000)
X_tsne = tsne.fit_transform(X_scaled)

plt.figure(figsize=(10, 8))
plt.scatter(X_tsne[:, 0], X_tsne[:, 1], alpha=0.7, c='coral', edgecolor='k')
plt.xlabel('t-SNE Dimension 1')
plt.ylabel('t-SNE Dimension 2')
plt.title('t-SNE Projection of Customer Purchasing Data')
plt.grid(True)
plt.show()
```

Confusion Matrix

Not Applicable

Interpretation

t-SNE is an unsupervised dimensionality reduction algorithm.

Since no ground-truth labels exist, a confusion matrix cannot be generated.

The algorithm preserves local neighborhood relationships and helps visualize hidden customer groups.

Graph:



✓ t-SNE Projection of Customer Purchasing Data

Graph Interpretation:

The t-SNE graph reveals several distinct customer clusters that were not clearly visible in the PCA visualization.

The algorithm successfully preserves nonlinear relationships between customers.

Different customer groups correspond to different purchasing behaviors.

Metrics:

Metric	Value
Accuracy	Not Applicable
Precision	Not Applicable
Recall	Not Applicable
F1-Score	Not Applicable
AUC-ROC	Not Applicable
MSE	Not Applicable
RMSE	Not Applicable
MAE	Not Applicable
R ²	Not Applicable

Algorithm 3: K-Means Clustering

Code:

```
k = 4
k-means = KMeans(n_clusters=k, random_state=42, n_init=10)
clusters = kmeans.fit_predict(X_scaled)

pca_df = pd.DataFrame(X_pca, columns=['PC1', 'PC2'])
pca_df['Cluster'] = clusters

plt.figure(figsize=(10, 8))
sns.scatterplot(data=pca_df, x='PC1', y='PC2', hue='Cluster', palette='Set2', alpha=0.8,
edgecolor='k')
plt.title('K-Means Clusters (k=4) visualized in PCA space')
plt.grid(True)
plt.legend(title='Cluster')
plt.show()

centroids = scaler.inverse_transform(kmeans.cluster_centers_)
centroid_df = pd.DataFrame(centroids, columns=df.columns)
```



```
centroid_df.index = [f'Cluster {i}' for i in range(k)]  
print("Cluster centroids (average purchase amounts):")  
print(centroid_df)
```

Confusion Matrix

Not Applicable

Interpretation:

K-Means is an unsupervised clustering algorithm.

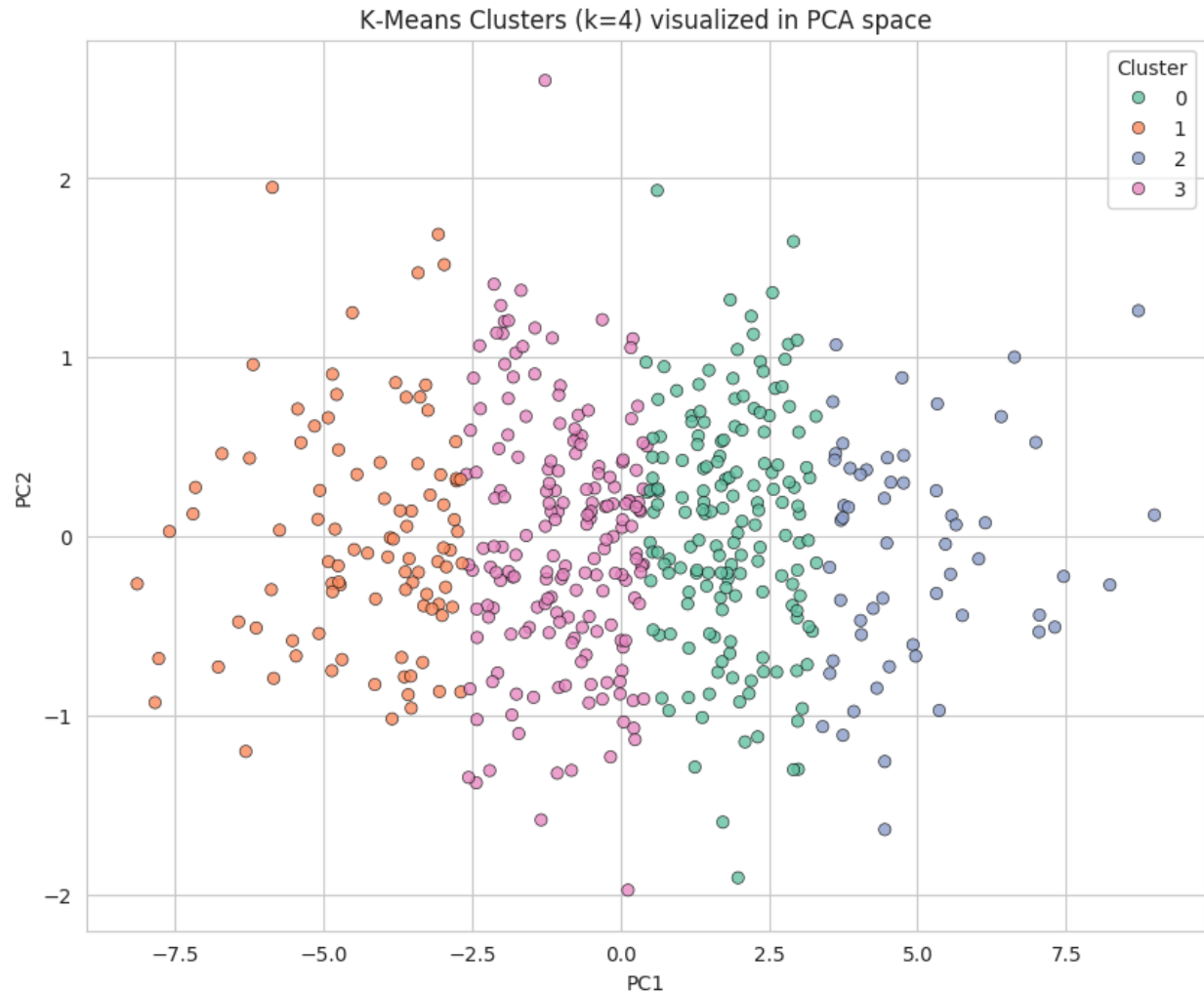
Since there are no predefined class labels, confusion matrix evaluation is not applicable.

The algorithm divides customers into four clusters according to their purchasing behavior.

Graph

Paste the following graph:

 K-Means Clusters (k = 4) Visualized in PCA Space



Graph Interpretation

The K-Means visualization clearly separates customers into four distinct purchasing groups.

Each cluster represents customers with similar spending behavior.

The centroid values help identify:

- High-spending customers
- Medium-spending customers
- Low-spending customers

These insights can be used for customer segmentation and personalized marketing.

Metrics

Metric	Value
Accuracy	Not Applicable
Precision	Not Applicable
Recall	Not Applicable
F1-Score	Not Applicable
AUC-ROC	Not Applicable
MSE	Not Applicable
RMSE	Not Applicable
MAE	Not Applicable
R ²	Not Applicable

5. Best Model Selection:

Best Model

K-Means Clustering with PCA Visualization

Reason:

The three algorithms were compared based on their ability to visualize customer purchasing behavior and identify meaningful customer groups.

- **PCA** efficiently reduced the dataset dimensions while preserving approximately **93.79%** of the total variance.
- **t-SNE** revealed hidden nonlinear patterns and showed clear customer groupings.
- **K-Means Clustering** successfully divided customers into four meaningful clusters based on purchasing behavior.

Among the three approaches, **K-Means Clustering with PCA Visualization** provided the most practical and interpretable customer segmentation. It identified distinct customer groups and generated cluster centroids that can be directly used for business decision-making, targeted marketing, and personalized recommendations.

6. Colab Link:

 [Customer Purchasing.ipynb](#)
